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Histidine-rich Glycoprotein Binds to Cell-surface Heparan Sulfate via Its N-terminal Domain following Zn^{2+} Chelation*

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Histidine-rich glycoprotein (HRG) is an α_2 -glycoprotein found in mammalian plasma at high concentrations ($\sim 150 \mu\text{g/ml}$) and is distinguished by its high content of histidine and proline. Structurally, HRG is a modular protein consisting of an N-terminal cystatin-like domain (N1N2), a central histidine-rich region (HRR) flanked by proline-rich sequences, and a C-terminal domain. HRG binds to cell surfaces and numerous ligands such as plasminogen, fibrinogen, thrombospondin, C1q, heparin, and IgG, suggesting that it may act as an adaptor protein either by targeting ligands to cell surfaces or by cross-linking soluble ligands. Despite the suggested functional importance of HRG, the cell-binding characteristics of the molecule are poorly defined. In this study, HRG was shown to bind to most cell lines in a Zn^{2+} -dependent manner, but failed to interact with the Chinese hamster ovary cell line pgsA-745, which lacks cell-surface glycosaminoglycans (GAGs). Subsequent treatment of GAG-positive Chinese hamster ovary cells with mammalian heparanase or bacterial heparinase III, but not chondroitinase ABC, abolished HRG binding. Furthermore, blocking studies with various GAG species indicated that only heparin was a potent inhibitor of HRG binding. These data suggest that heparan sulfate is the predominate cell-surface ligand for HRG and that mammalian heparanase is a potential regulator of HRG binding. Using recombinant forms of full-length HRG and the N-terminal N1N2 domain, it was shown that the N1N2 domain bound specifically to immobilized heparin and cell-surface heparan sulfate. In contrast, synthetic peptides corresponding to the Zn^{2+} -binding HRR of HRG did not interact with cells. Furthermore, the binding of full-length HRG, but not the N1N2 domain, was greatly potentiated by physiological concentrations of Zn^{2+} . Based on these data, we propose that the N1N2 domain binds to cell-surface heparan sulfate and that the interaction of Zn^{2+} with the HRR can indirectly enhance cell-surface binding.

Histidine-rich glycoprotein (HRG)¹ is an ~ 75 -kDa single polypeptide chain α_2 -plasma glycoprotein synthesized in the

liver and found in the plasma of most vertebrates at a relatively high concentration of 100 – $200 \mu\text{g/ml}$ ($\sim 2 \mu\text{M}$). The most distinctive feature of HRG arises from its high content of histidine and proline residues, which each account for $\sim 13\%$ of the total amino acids (1). HRG is predicted to be a multidomain protein consisting of two cystatin-like domains at the N terminus (termed N1 and N2), a central histidine-rich region (HRR) flanked by two proline-rich sequences, and a C-terminal domain. HRG is known to bind a variety of ligands, including heme and Zn^{2+} (2–4), plasminogen (5, 6), fibrinogen (7), thrombospondin and IgG (8, 9), C1q (9), and heparin (10–12), and can interact with various cell-surface receptors, including Fc γ receptors (13, 14) and an undefined T-cell receptor (15). Thus, HRG may potentially regulate numerous biological processes such as hemostasis, fibrinolysis, thrombosis, angiogenesis, leukocyte migration, and cancer metastasis.

Despite considerable interest in HRG, many of the fundamental characteristics regarding HRG cell-surface binding remain undefined. For example, despite the identification of numerous soluble ligands for HRG, the cell-surface ligands are not well characterized. Indirect evidence suggests that negatively charged glycosaminoglycans (GAGs) may mediate HRG cell-surface binding, as heparin is a potent inhibitor of HRG binding to cells (16, 17). Zn^{2+} is also known to interact with HRG, most probably with the proposed metal chelation sites located within the HRR (2–4), with this interaction enhancing the binding of HRG to cells (18). Similarly, the location of the cell surface-binding domain within HRG is unclear. The ability of Zn^{2+} to enhance cell-surface binding and the assumption that heparin binds to the HRR and inhibits binding have led to the hypothesis that the HRR interacts with cells surfaces (10, 12, 16, 19). In contrast, it has also been suggested that a heparin-binding sequence may be located in the N1N2 domain, which raises the possibility that this domain of HRG may interact with cells (17, 20, 21).

In this study, we have characterized the cell surface-binding properties of HRG using recombinant full-length HRG and the recombinant N1N2 domain together with various approaches to remove GAGs from cell surfaces. We provide clear evidence that heparan sulfate is the dominant cell-surface ligand for HRG, with the interaction being mediated through the N1N2 domain of HRG. Indeed, enzyme-linked immunosorbent assay (ELISA) studies confirmed that the N1N2 domain specifically binds to immobilized heparin with comparable affinity to full-length HRG. Furthermore, cell-surface binding of full-length HRG, but not the N1N2 domain, was greatly potentiated by the presence of physiological concentrations of Zn^{2+} . Based on

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¹ The abbreviations used are: HRG, histidine-rich glycoprotein; HRR,

histidine-rich region; GAG, glycosaminoglycan; ELISA, enzyme-linked immunosorbent assay; Ni-NTA, nickel-nitrilotriacetic acid; PBS, phosphate-buffered saline; mAb, monoclonal antibody; BSA, bovine serum albumin; CHO, Chinese hamster ovary.

these data, we propose a model whereby HRG binds to cell-surface heparan sulfate via its N1N2 domain with low affinity, which is enhanced following Zn^{2+} binding to the HRR. Thus, HRG may play an important physiological and/or pathological role by binding to cell surfaces in local environments that contain high levels of Zn^{2+} such as sites of inflammation or during tumor metastasis and angiogenesis.

EXPERIMENTAL PROCEDURES

Cell Lines—B16F1, COS-7, MT4, HT1080, and Jurkat cells were cultured in RPMI 1640 medium (Invitrogen) supplemented with 10% fetal calf serum. CHO-K1 and pgsA-745 cells were cultured in Dulbecco's modified Eagle's medium (Invitrogen) supplemented with 21 $\mu\text{g}/\text{ml}$ L-proline and 10% fetal calf serum. Mammalian cell lines were incubated at 37 °C in a humidified atmosphere containing 5% CO_2 . The *Spodoptera frugiperda*-derived insect cell line Sf9 was cultured in Sf-900 II serum-free medium (Invitrogen) at 27 °C.

Purification of HRG—Native human HRG was purified from fresh human plasma as described previously (22). Briefly, a phosphocellulose column was equilibrated with loading buffer (0.5 M NaCl, 10 mM sodium phosphate, and 1 mM EDTA (pH 6.8)) for 24 h. Fresh human plasma was provided by Red Cross House of Canberra Hospital (Canberra, Australia) and mixed with NaCl and EDTA to final concentrations similar to those of the loading buffer and with the protease inhibitors aprotinin (2 $\mu\text{g}/\text{ml}$), phenylmethylsulfonyl fluoride (100 $\mu\text{g}/\text{ml}$), and 4-(2-aminoethyl)benzenesulfonyl fluoride hydrochloride (100 $\mu\text{g}/\text{ml}$). The plasma was passed through the equilibrated column; unbound protein was removed by extensive washing with loading buffer; and bound HRG was eluted with 2.0 M NaCl, 10 mM sodium phosphate, and 1 mM EDTA (pH 6.8).

Synthesis of L1–L5 Peptides—The HRR within human HRG is composed of 12 tandem repeats, with the dominant consensus repeat sequence being the 5-amino acid motif GHHPH. Synthetic peptides composed of one to five repeats of this 5-amino acid motif (termed L1–L5, respectively) were produced. L1–L3 peptides biotinylated at the N terminus were also prepared. The L1–L5 peptides were provided as a kind gift by Dr. Joe Altin (School of Biochemistry and Molecular Biology, Australian National University).

Plasmid Constructs—Using the pUC9-HRG plasmid construct as a template (1), PCR with oligonucleotides ALJ-1 and ALJ-5 was carried out to amplify a cDNA encoding full-length human HRG comprising the leader, first cystatin (N1) domain, second cystatin (N2) domain, HRR, and C-terminal domain coding regions. The N1N2 coding region cDNA was generated by amplification with oligonucleotides ALJ-1 and ALJ-3. N1N2 was engineered to contain a hexahistidine tag at the 3'-end of the molecule through inclusion of six tandem histidine codons in oligonucleotide ALJ-3. The oligonucleotide primer sequences used are as follows: ALJ-1, 5'-TTGAATTCTAAAGGGATGGTTTAAACAAAATG-3'; ALJ-3, 5'-AAGGTACCTTAGTGATGGTGATGGTGATGCTGAGGGTCGAAGACTTCAC-3'; and ALJ-5, 5'-AAGGTACCTTATTTTGGAAATGTATGTGTAAAAAC-3'. The amplified HRG and N1N2 cDNAs were cloned into the EcoRI/KpnI sites in the donor vector construct pFastBac (Invitrogen), generating the pFastBac-HRG and pFastBac-N1N2 constructs, respectively. The nucleotide integrity of each clone was confirmed by automated sequencing using an Applied Biosystems 3730 DNA analyzer.

Transfections and Recombinant Protein Production Using the Baculovirus Expression System—Recombinant proteins of full-length human HRG and the N2N2 domain of human HRG were produced using the "Bac-to-Bac" baculovirus expression system (Invitrogen). Recombinant bacmid constructs were generated using the pFastBac-HRG or pFastBac-N1N2 construct according to the manufacturer's instructions. Briefly, Sf9 cells were transfected using Cellfectin reagent (Invitrogen), with 1–2 μg of recombinant bacmid DNA being transfected per 9×10^5 cells for 5 h at 27 °C. Supernatant containing recombinant baculovirus was harvested 72 h post-transfection and was amplified for 3–4 days at 27 °C by infecting Sf9 cells with a multiplicity of infection of ~0.01–0.1. Typically, recombinant baculovirus was amplified three times before being used to produce recombinant protein. Harvested Sf9 supernatant containing recombinant protein was purified using nickel-nitrilotriacetic acid (Ni-NTA)-agarose (QIAGEN Inc., Hilden, Germany). Recombinant full-length HRG and hexahistidine-tagged N1N2 were eluted from Ni-NTA-agarose with 200 mM cold imidazole.

Western Blotting—Recombinant proteins were boiled for 10 min in 20 μl of SDS reducing sample buffer (125 mM Tris-HCl (pH 6.8), 20% glycerol, 10% dithiothreitol, and 4% SDS) and then subjected to electrophoresis on a 4–20% (w/v) gradient precast polyacrylamide minigel

(Gradipore, Sydney, Australia). Proteins were transferred electrophoretically using a Mini-Protein II apparatus (Bio-Rad) onto a nitrocellulose membrane using a transfer buffer containing 48 mM Tris and 39 mM glycine in 20% (v/v) methanol. The membrane was blocked overnight with 5% (w/v) skim milk powder diluted in phosphate-buffered saline (PBS). Full-length HRG and the N1N2 domain were detected using the HRG-specific monoclonal antibody (mAb) HRG-4 (AGEN, Brisbane, Australia) and by chemiluminescence using ECL Western blotting detection reagents (Amersham Biosciences, Buckinghamshire, United Kingdom).

Immunofluorescence Flow Cytometry—Cell lines were analyzed for HRG binding or cell-surface heparan sulfate expression by immunofluorescence flow cytometry. Typically, plasma-purified or recombinant HRG or the recombinant N1N2 domain (100 $\mu\text{g}/\text{ml}$) was added to 5×10^5 cells in PBS and 0.1% bovine serum albumin (BSA) with or without 20 μM Zn^{2+} for 60 min at 4 °C and washed three times with PBS and 0.1% BSA. Cell-bound HRG or N1N2 was detected using the HRG-specific mAb HRG-4, and cell-surface heparan sulfate was detected by mAb F58-10E4 (Seikagaku Corp., Tokyo, Japan), followed by secondary detection with fluorescein isothiocyanate-labeled sheep anti-mouse Ig (Amrad Biotech, Melbourne, Australia). Cells were analyzed by immunofluorescence flow cytometry using an LSR flow cytometer (BD Biosciences), with forward scatter, side scatter, and FL-1 data being collected. Flow cytometry data were analyzed using CellQuest Pro software (BD Biosciences). Each treatment condition was typically repeated in triplicate, and each experiment was repeated two to three times unless stated otherwise. In some experiments, cells were treated with mammalian heparanase (2 units/ml), bacterial heparinase III (2 units/ml; Sigma), or chondroitinase ABC (2 units/ml; Sigma) diluted in PBS and 0.1% BSA for 2 h at 37 °C. In other experiments, human HRG (100 $\mu\text{g}/\text{ml}$) was co-incubated with different GAGs (0.5–100 $\mu\text{g}/\text{ml}$), including bovine lung heparin (3.1, 4.5, 10.6, 12.5, and 16.7 kDa) and chondroitin sulfates A, B, C, and E (Sigma). Also, in some cases, cells were incubated with the biotinylated L1–L3 peptides (100 μM), with peptide binding to the cells being detected by R-phycoerythrin-conjugated streptavidin (Caltag Laboratories, Burlingame, CA).

ELISAs—ELISAs were performed by coating 96-well polyvinyl chloride microtiter plastic plates (Dynex Technologies Inc., Chantilly, VA) overnight at 4 °C with recombinant full-length HRG or N1N2 (50 μl /well, ~5 $\mu\text{g}/\text{ml}$) in 0.05 M $\text{Na}_2\text{CO}_3/\text{NaHCO}_3$ buffer (pH 9.6) (Sigma) or with streptavidin (50 μl /well, 10 $\mu\text{g}/\text{ml}$; Sigma) diluted in PBS. Plates were then washed with PBS and 0.02% Tween 20 and blocked for 120 min at room temperature with 3% (w/v) BSA diluted in PBS. In some experiments, biotinylated heparin (10 $\mu\text{g}/\text{ml}$) diluted in PBS and 1% BSA was added to the streptavidin-coated plates for 60 min at room temperature before addition of HRG or the N1N2 domain (50 μl /well, ~0.1–100 nM) diluted in PBS in the absence or presence of 20 μM Zn^{2+} and/or 1 mM EDTA. Bound HRG and N1N2 were detected using the HRG-specific mAb HRG-4, followed by secondary antibody detection with horseradish peroxidase-conjugated sheep anti-mouse Ig (Amrad Biotech). Plate-bound peroxidase was detected using 2,2'-azinobis(3-ethylbenzothiazoline-6-sulfonic acid) peroxidase substrate (Kirkegaard & Perry Laboratories Inc., Gaithersburg, MD) and measuring the absorbance at 405 nm (reference wavelength of 490 nm) on a Thermomax microplate reader. Data were analyzed using SoftMax Pro software (Molecular Devices, Sunnyvale, CA).

RESULTS

HRG Binding to Cell Surfaces Is Zn^{2+} -dependent—Previous studies have suggested that the binding of HRG to cell surfaces is highly Zn^{2+} -dependent, with physiological concentrations of Zn^{2+} (~20 μM) being particularly efficacious (18). Initial experiments used immunofluorescence flow cytometry to analyze the effects of 20 μM Zn^{2+} on the binding of plasma-derived HRG to six different mammalian cell lines with widely differing tissue and species origins, *viz.* mouse melanoma cells (B16F1), human T-cells (MT4 and Jurkat), monkey kidney fibroblasts (COS-7), human fibrosarcoma cells (HT1080), and human umbilical vein endothelial cells (Fig. 1A). HRG bound to five of the six lines tested in a highly Zn^{2+} -dependent manner, with the exception of human umbilical vein endothelial cells, which exhibited negligible HRG binding in both the presence and absence of Zn^{2+} .

Human HRG Interacts with Cell-surface Heparan Sulfate—It has been postulated previously that cell-surface GAGs such as heparan sulfate are able to interact with HRG (16, 17).

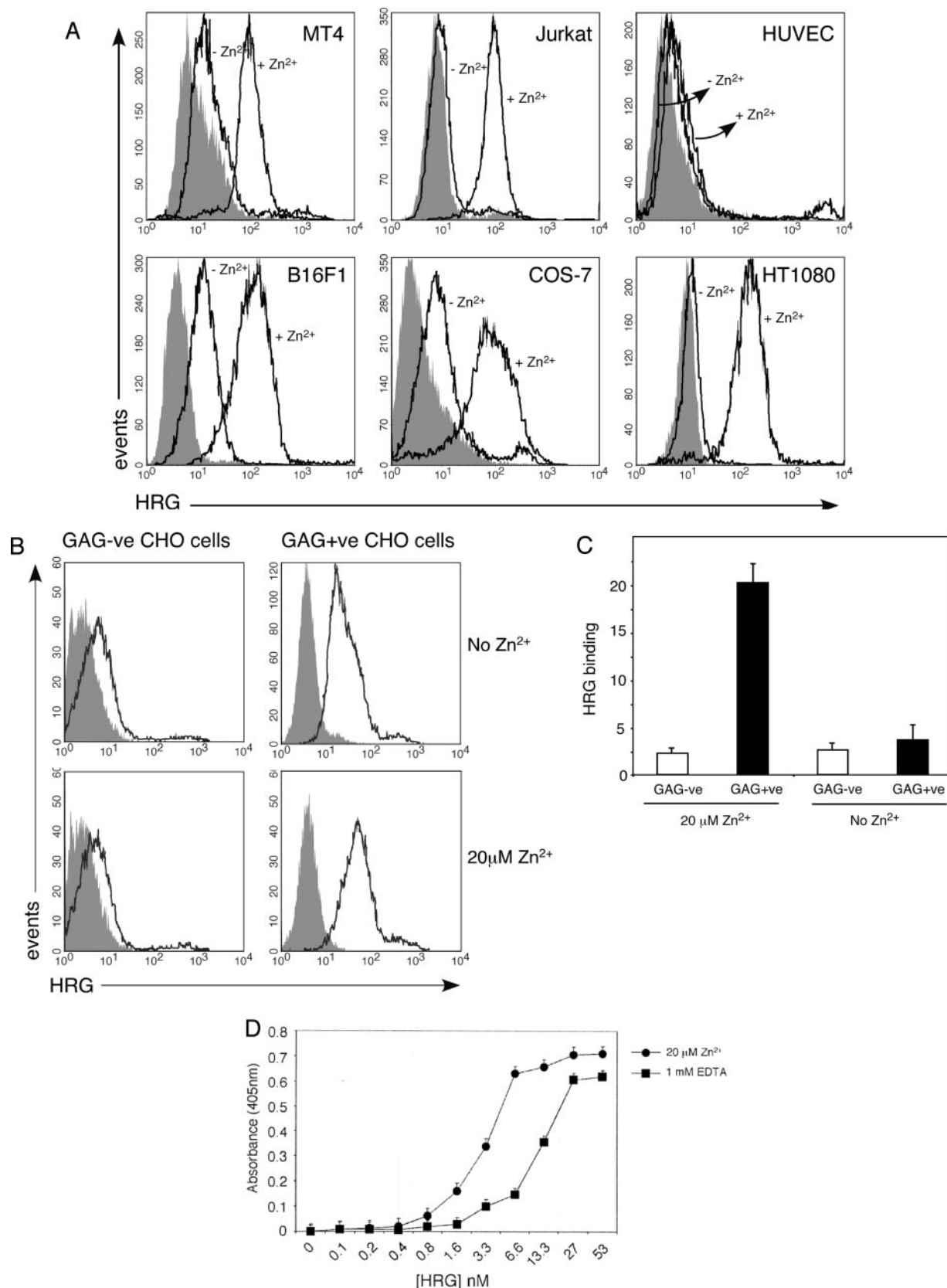


FIG. 1. Role of cell-surface GAGs and Zn^{2+} in the binding of human HRG to cell surfaces. A, various cell types were incubated with plasma-derived HRG (100 $\mu g/ml$) in the presence or absence of 20 μM Zn^{2+} and analyzed for HRG binding by immunofluorescence flow cytometry. Filled histograms represent background fluorescence, and empty histograms represent HRG binding as detected by the HRG-specific mAb HRG-4. HUVEC, human umbilical vein endothelial cells. B, CHO cell lines that express cell-surface GAGs (GAG+ve CHO) or that lack cell-surface GAGs (GAG-ve CHO) were incubated with 100 $\mu g/ml$ plasma-derived HRG in the presence or absence of 20 μM Zn^{2+} and then analyzed for HRG binding by immunofluorescence flow cytometry. C, the numerical values show HRG binding as -fold increase above background for GAG-ve CHO and GAG+ve CHO cells with or without 20 μM Zn^{2+} . Error bars represent S.E. ($n = 3$). D, the results from ELISA show the binding of plasma-derived full-length HRG (0.1–53 nM), diluted in PBS and 1% BSA containing 20 μM Zn^{2+} or 1 mM EDTA, to plastic immobilized heparin. HRG binding was detected using the HRG-specific mAb HRG-4. Error bars represent S.E. ($n = 3$).

To test this hypothesis directly, advantage was taken of Chinese hamster ovary (CHO) cell lines that either express cell-surface GAGs (CHO-K1) or lack cell-surface GAGs (pgsA-745) due to a deficiency in xylosyltransferase (23, 24). We found that GAG-expressing (GAG+ve) CHO cells bound HRG in a Zn^{2+} -dependent manner, but that GAG-deficient (GAG-ve) CHO cells did not bind HRG in either the presence or absence of $20 \mu\text{M}$ Zn^{2+} (Fig. 1, *B* and *C*). These results indicate that HRG binds to cell-surface GAGs and that this interaction is enhanced by physiological concentrations of Zn^{2+} ($20 \mu\text{M}$). Previous studies have reported that HRG interacts with the GAG heparin (10–12). Thus, ELISA studies were carried out to analyze the effect of Zn^{2+} on the binding of full-length HRG to plastic immobilized heparin. Consistent with previous results (16), HRG interacted with immobilized heparin, and this interaction was enhanced ~4-fold in the presence of $20 \mu\text{M}$ Zn^{2+} (Fig. 1*D*). Cell-surface GAGs are composed of a mixture of heparan sulfate, chondroitin sulfates A and C, dermatan sulfate (chondroitin sulfate B), and hyaluronic acid (25). To define the specific cell-surface GAGs that interact with HRG, we enzymatically removed heparan sulfate and chondroitin sulfates A, B, and C from GAG+ve CHO cells. Initially, using a heparan sulfate-specific mAb and immunofluorescence flow cytometry, we verified that GAG+ve CHO cells express high levels of cell-surface heparan sulfate and that GAG-ve CHO cells do not express heparan sulfate (Fig. 2*A*). The enzymatic activity of mammalian heparanase, bacterial heparinase III, and chondroitinase ABC was verified by demonstrating by fast protein liquid chromatography that the enzymes could cleave heparan sulfate or chondroitin 6-sulfate chains, respectively, into smaller fragments (data not shown). GAG+ve CHO cells that were treated with either mammalian heparanase or bacterial heparinase III were completely depleted of surface heparan sulfate (Fig. 2, *B* and *C*). Subsequently, HRG binding to these mammalian heparanase- and heparinase III-treated cells was found to be markedly reduced (~85–90%) (Fig. 2, *D* and *E*). In contrast, chondroitinase ABC treatment had no effect on HRG binding (Fig. 2, *D* and *E*). These results indicate that HRG binds specifically to cell-surface heparan sulfate and not chondroitin sulfates and that heparan sulfate is the principal cell-surface receptor for HRG on cells. Results from additional binding inhibition experiments carried out with a range of soluble GAGs were consistent with heparan sulfate being the GAG receptor for HRG on cells. Thus, various sized fragments of bovine lung heparin (4.5–16.7-kDa preparations) were potent inhibitors of the interaction of HRG with GAG+ve CHO cells, although very low molecular mass heparin (3.5 kDa) was a relatively ineffectual inhibitor (Table I). In contrast, chondroitin sulfates A, B, C, and E were totally inactive as inhibitors (Table I).

The HRR of HRG Does Not Bind to Cells—It has been suggested previously that the HRR of HRG interacts with heparin/heparan sulfate (12, 16, 19). To test this hypothesis, peptides corresponding to the histidine repeat sequence within the HRR of HRG were synthesized and used in binding studies. The peptides comprised one (L1), two (L2), three (L3), four (L4), or five (L5) repeats of the dominant pentapeptide motif GHHPH that is tandemly repeated in the HRR of human HRG. Immunofluorescence flow cytometry analysis indicated that biotinylated preparations of the L1–L3 peptides did not bind to GAG+ve CHO cells in either the presence or absence of $20 \mu\text{M}$ Zn^{2+} (Fig. 3*A*), with R-phycoerythrin-conjugated streptavidin being used to monitor peptide binding. Furthermore, HRG blocking experiments revealed that high concentrations ($100 \mu\text{M}$) of the L1–L5 peptides failed to inhibit full-length HRG binding to GAG-expressing CHO cells and B16F1 melanoma

cells (Fig. 3*B*). These data contrast with previous indirect evidence that the HRR binds heparan sulfate (12, 16, 19) and imply that the cell surface-binding domain of HRG may be located in another region of the molecule.

Production of Recombinant Full-length Human HRG and the N-terminal N1N2 Domain—Earlier sequence homology studies suggested that the N1N2 domain of HRG contains a heparin-binding motif and may represent the region of HRG that interacts with cell-surface heparan sulfate. To test this possibility, we produced recombinant full-length human HRG and the N-terminal fragment of HRG, N1N2 (Fig. 4*A*), in insect cells using a baculovirus expression system. A hexahistidine tag was engineered onto the C terminus of the N1N2 domain to allow purification of the recombinant protein by Ni-NTA chelation chromatography. Recombinant full-length HRG was made untagged, as it contains sufficient histidine residues in the HRR to allow Ni-NTA-agarose affinity purification. Western blot analysis of purified recombinant full-length HRG showed a single band at the anticipated molecular mass of 75 kDa, and the N1N2 preparation showed a prominent band at the appropriate molecular mass of 35 kDa using the HRG-specific mAb HRG-4 (Fig. 4*B*). Proteins present in an Sf9 cell culture supernatant that bound to Ni-NTA-agarose were included as a negative control. ELISA studies (Fig. 4*C*) indicated that full-length HRG and the N1N2 domain were both recognized by the HRG-specific mAb HRG-4, demonstrating that the epitope recognized by HRG-4 is located within the N1N2 domain of HRG.

The N1N2 Domain of HRG Binds to Cells—ELISA studies showed that the recombinant N1N2 domain of HRG bound to plastic immobilized heparin with similar affinity to recombinant full-length HRG in the presence of $20 \mu\text{M}$ Zn^{2+} (Fig. 5*A*). Immunofluorescence flow cytometry studies were performed to determine whether the N1N2 domain binds to cells. First, it was established that recombinant full-length HRG produced in insect cells bound to GAG+ve CHO cells. Indeed, recombinant full-length HRG, as with plasma-derived HRG, bound to GAG-expressing CHO cells in a Zn^{2+} -inducible manner (Fig. 5, *B* and *D*), and the binding was totally inhibited by $100 \mu\text{g/ml}$ 12.5-kDa heparin (Fig. 5*B*). As expected, recombinant full-length HRG did not bind to GAG-deficient CHO cells (Fig. 5, *B* and *D*). Similarly, N1N2 bound to GAG+ve CHO cells in the presence of $20 \mu\text{M}$ Zn^{2+} and, interestingly, also bound equally well to these cells in the absence of $20 \mu\text{M}$ Zn^{2+} (Fig. 5, *C* and *D*). Similar to full-length HRG, the binding of the N1N2 domain to GAG-expressing CHO cells was also inhibited by $100 \mu\text{g/ml}$ 12.5-kDa heparin, suggesting that the N1N2 domain interacts with the same putative cell-surface receptor as full-length HRG, *viz.* heparan sulfate (Fig. 5*C*). Consistent with this observation, the N1N2 domain did not bind to GAG-deficient CHO cells (Fig. 5, *C* and *D*).

DISCUSSION

Despite numerous studies suggesting that HRG can interact with cell surfaces, the basic cell-binding characteristics of the molecule remain poorly defined. In this study, we have demonstrated that human HRG at physiological concentrations (~100 $\mu\text{g/ml}$) bound to cell-surface heparan sulfate via its N-terminal domain (N1N2) and that binding was greatly potentiated by the presence of physiological concentrations of Zn^{2+} (~20 μM). Similarly, it was shown that full-length HRG bound to immobilized heparin, an interaction that was enhanced ~4-fold in the presence of $20 \mu\text{M}$ Zn^{2+} , and that the N1N2 domain of HRG bound to immobilized heparin with a comparable affinity to full-length HRG.

It has been postulated previously that HRG binds to negatively charged GAGs on cell surfaces (17, 26); however, no direct evidence was provided to support this claim. Our results

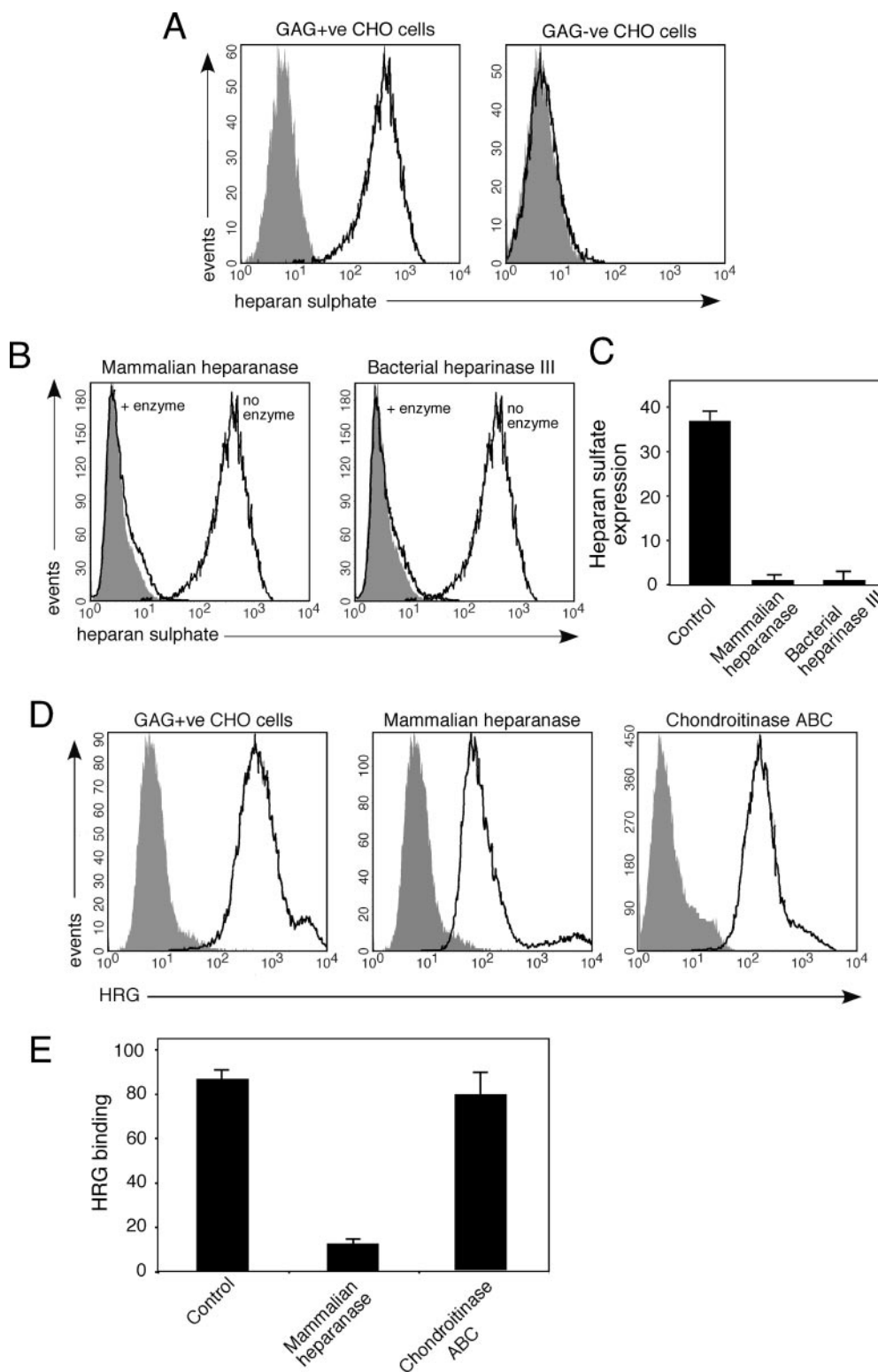


FIG. 2. Human HRG interacts with cell-surface heparan sulfate and not chondroitin sulfate. *A*, cell-surface expression of heparan sulfate by the GAG-expressing (GAG+ve) and GAG-deficient (GAG-ve) CHO cell lines assessed by immunofluorescence flow cytometry using a heparan sulfate-specific mAb. *B*, effect of mammalian heparanase or bacterial heparinase III treatment on heparan sulfate expression by the GAG-expressing CHO cell line. *C*, -fold change in heparan sulfate expression in GAG-expressing CHO cells following mammalian heparanase or bacterial heparinase III treatment. Data are means $\pm$ S.E. of three determinations. *D* and *E*, effect of mammalian heparanase and chondroitinase ABC treatment on HRG binding to GAG+ve CHO cells. *D* shows representative flow cytometry histograms (filled histograms, background fluorescence; empty histograms, heparan sulfate expression or HRG binding as detected by the appropriate mAb), and *E* depicts HRG binding (-fold increase in binding above background) following the different treatments. Data are means $\pm$ S.E. of three determinations. Cells were incubated with 100 μ g/ml plasma-derived HRG in the presence of 20 μ M Zn^{2+} .

reveal that HRG cannot bind to the CHO cell line pgsA-745, which lacks cell-surface GAGs, although HRG readily binds to the parent cell line, CHO-K1, which is not GAG-deficient. Fur-

thermore, treatment of the CHO-K1 cell line with mammalian heparanase or bacterial heparinase III, processes that remove cell-surface heparan sulfate, essentially abolished HRG bind-

TABLE I
Ability of different GAGs to inhibit cell-surface binding of HRG to CHO-K1 cells

GAG	IC ₅₀ ^a μg/ml
Heparin	
3.1 kDa	50
4.5 kDa	6
10.6 kDa	3
12.5 kDa	3
16.7 kDa	3
Chondroitin sulfate A	>100
Chondroitin sulfate B	>100
Chondroitin sulfate C	>100
Chondroitin sulfate E	>100

^a Concentration of GAG that inhibited HRG binding to GAG+ve CHO cells by 50% as measured by immunofluorescence flow cytometry.

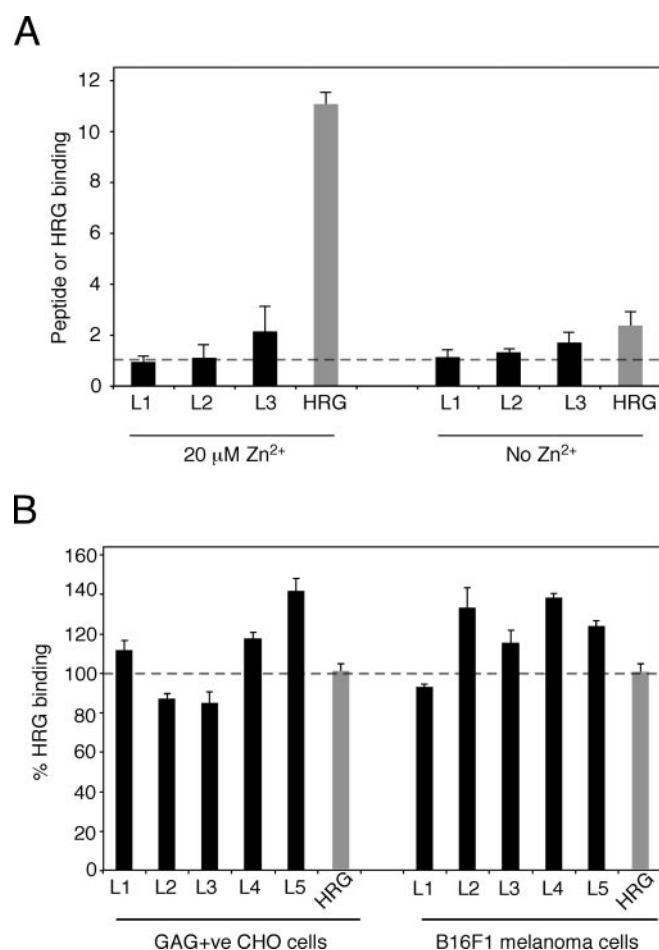


FIG. 3. Peptides corresponding to the histidine-rich repeat sequence of human HRG do not bind to GAG-expressing CHO cells and do not block HRG binding to GAG-expressing CHO cells. A, GAG+ve CHO cells were incubated in the presence or absence of 20 μM Zn²⁺ with biotinylated peptides that correspond to one (L1), two (L2), or three (L3) repeats of the consensus histidine pentapeptide of the HRR of HRG (GHHPH) (100 μM), with peptide binding being detected using R-phycoerythrin-conjugated streptavidin by immunofluorescence flow cytometry. Results are shown as -fold increase in peptide binding above background, with a value of 1 representing background binding (shown as a dashed line). Binding of HRG (100 μg/ml) as detected by the HRG-specific mAb HRG-4 is presented as a control. B, GAG+ve CHO cells and B16F1 melanoma cells were incubated with HRG (100 μg/ml) and 20 μM Zn²⁺ in the presence or absence of peptides corresponding to one to five repeats of the consensus histidine pentapeptide (L1–L5; 100 μM) and then analyzed for HRG binding using the HRG-specific mAb HRG-4 by immunofluorescence flow cytometry. Results are shown as percent HRG binding compared with control HRG binding (dashed line). Error bars represent S.E. (n = 3).

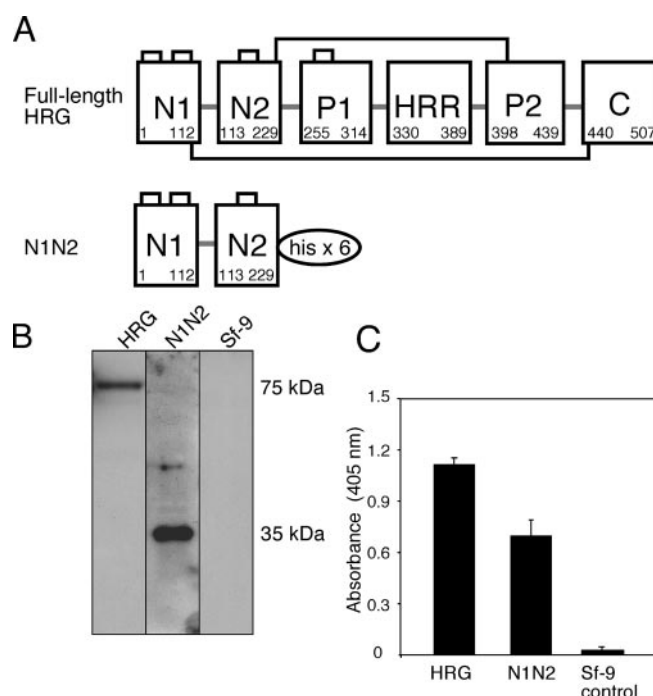


FIG. 4. Production of recombinant full-length human HRG and the N-terminal domain of HRG (N1N2). A, schematic representation of recombinant full-length HRG and the N-terminal fragment of HRG (N1N2). The N1N2 domain contains a hexahistidine tag engineered onto the C terminus of the molecule. Inter- and intradomain disulfide bonds are represented by thick black lines. Recombinant proteins were produced by baculovirus expression in the Sf9 insect cell line. P1 and P2, proline-rich sequences. B, Western blot analysis of Ni-NTA-agarose-purified recombinant HRG and N1N2 on reducing SDS-polyacrylamide gels. Recombinant HRG (75 kDa) and the N1N2 domain (35 kDa) were detected by the HRG-specific mAb HRG-4. An Sf9 cell culture supernatant was purified using Ni-NTA-agarose and was run as a negative control. C, ELISA showing that plastic immobilized recombinant HRG and N1N2 bound the HRG-specific mAb HRG-4, with Sf9 proteins, as described for B, included as a control. Error bars represent S.E. (n = 3).

ing, whereas chondroitinase ABC treatment had no effect. Combined with the observation that heparin was the only GAG tested that inhibited HRG binding to cells, these data support the conclusion that heparan sulfate is an important cell-surface ligand for HRG. HRG has been reported to interact with other specific cell-surface receptors such as the Fcγ receptor (13, 14, 27) and an undefined T-cell receptor (15); however, ubiquitously expressed heparan sulfate is likely to be the predominant HRG receptor on cells. Of particular interest was the finding that treatment of cells with mammalian heparanase resulted in the complete removal of cell-surface heparan sulfate, with a resultant reduction in HRG binding. This result implies that the local production of heparanase may provide an endogenous mechanism for regulating the interaction of HRG with cell surfaces.

We also conducted studies to define the domain of HRG that binds to cells. Our results contrast with previous circumstantial evidence that suggests that HRG binds to cells via its HRR and that heparin binds to the HRR (10, 12, 16, 19). These previous studies relied on the observation that Zn²⁺, which binds to chelation sites in the HRR, regulates cell-surface binding, but no direct evidence has been published demonstrating that the HRR binds to cells. To resolve the issue, we initially examined whether synthetic peptides representing multiple repeats (one to five) of the dominant pentapeptide sequence in the HRR of human HRG interacted with cells. Contrary to earlier predictions, it was found that all five synthetic peptides were unable to inhibit HRG binding to cells, even when used at

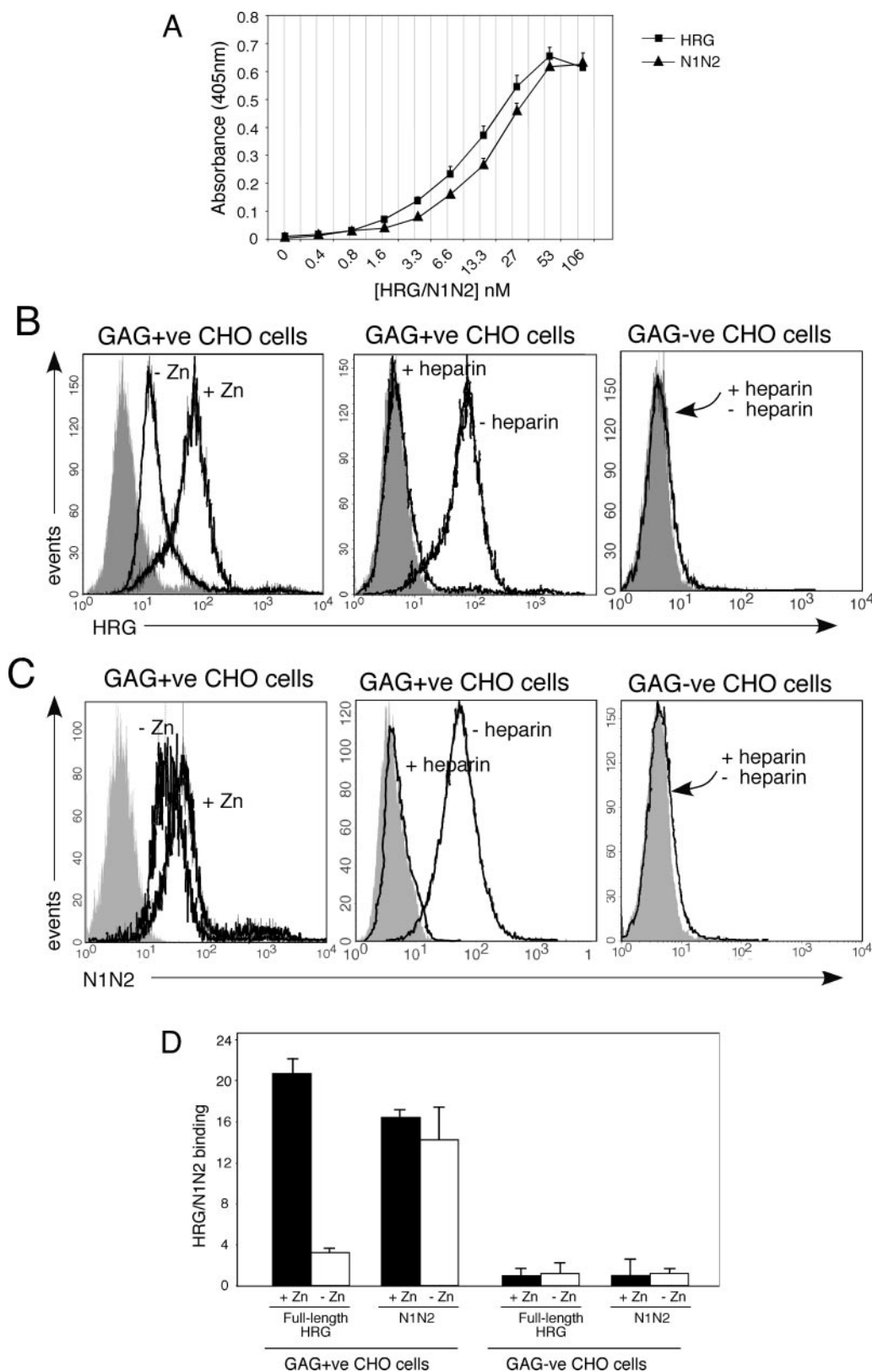


FIG. 5. Ability of recombinant human HRG and the N1N2 domain of HRG to bind to cell surfaces. *A*, ELISA depicting the ability of recombinant full-length HRG and the N1N2 domain of HRG (0.1–53 nM), diluted in PBS, 1% BSA, and 20 μM Zn^{2+} , to bind to plastic immobilized heparin. HRG and N1N2 binding was detected using the HRG specific mAb HRG-4. Error bars represent S.E. ($n = 3$). *B*, binding of recombinant full-length HRG (50 $\mu\text{g}/\text{ml}$) to GAG+ve CHO cells in the presence or absence of 20 μM Zn^{2+} (left panel), to GAG+ve CHO cells in the presence or absence of 100 $\mu\text{g}/\text{ml}$ ~12.5-kDa bovine lung heparin and 20 μM Zn^{2+} (center panel), or to GAG-ve CHO cells in the presence of 20 μM Zn^{2+} with or without heparin (right panel) as assessed by immunofluorescence flow cytometry. Filled histograms represent background fluorescence. *C*, representative flow cytometry histograms of binding of the N1N2 domain (50 $\mu\text{g}/\text{ml}$) to GAG+ve CHO and GAG-ve CHO cells as described for *B*. *D*, quantitative comparison of the Zn^{2+} dependence of binding of recombinant full-length HRG and the N1N2 domain of HRG to GAG+ve CHO and GAG-ve CHO cells. HRG/N1N2 binding is expressed as -fold increase in binding relative to background autofluorescence. Error bars represent S.E. ($n = 3$).

a 50-fold molar excess, and biotinylated versions of three of the peptides (one to three pentapeptide repeats) also did not bind to CHO-K1 cells. Recent studies in our laboratory also indicate that these same synthetic peptides containing multiple pentapeptide repeats exhibit anti-angiogenic activity *in vitro*,² which is in agreement with studies by Simantov *et al.* (28) and Olsson *et al.* (29), suggesting that the HRR plays an important role in targeting angiogenesis and not in mediating the binding of HRG to cells. Because it has also been hypothesized that the N1N2 domain of HRG contains a heparin-binding site (30), we produced recombinant forms of both full-length HRG and the N-terminal N1N2 domain of HRG to directly investigate the cell surface-binding properties of the N1N2 domain. Our results clearly indicate that the N1N2 domain binds to immobilized heparin and to cells in a heparin-inhibitable manner and that, unlike full-length HRG, this binding is not dependent on Zn^{2+} .

We analyzed the sequence of the N-terminal domain of HRG and were unable to identify any classic heparin-binding motifs (31). However, not all heparin-binding proteins exhibit classic heparin-binding motifs, as in some cases, the protein conformation brings together basic residues that are actually distant in the linear sequence. Thus, the spatial orientation of basic residues rather than sequence proximity remains an important factor in determining heparin-binding capacity. Antithrombin III represents one such heparin-binding protein that does not contain a classic heparin-binding motif, where it appears that both linearly contiguous basic heparin-binding residues as well as remote basic residues are appropriately positioned in the structure of the protein to bind heparin (32). The heparin-binding domain of antithrombin III is located within its N-terminal domain, and interestingly, HRG shares ~40% sequence identity with this region of antithrombin III (20, 21, 33). The heparin-binding domain of antithrombin III has been extensively studied, and Arg⁴⁶, Arg⁴⁷, Lys¹¹⁴, and Lys¹²⁵ have been identified as key basic residues that bind to negatively charged heparin (30, 34). Interestingly, a number of these residues are also conserved in HRG. Arg⁴⁶ and Arg⁴⁷ are conserved in human, mouse, rat, and rabbit HRGs; Lys¹²⁵ is conserved in human, mouse, and rat HRGs; and Lys¹¹⁴ is substituted with a conserved basic residue (Arg) in human HRG. Collectively, the sequence alignment between HRG and antithrombin III suggests that Lys²², Arg²³, Arg⁷⁷, Arg⁷⁸, Arg¹³⁵, and Lys¹⁴⁶ in human HRG could possibly constitute heparin-binding residues. However, we can only speculate that these basic amino acids constitute the heparin-binding region within HRG, and clearly, further work is needed to define the specific heparin/heparan sulfate-binding residues within the N1N2 domain of HRG.

One of the most intriguing aspects of this study was the observation that the binding of full-length HRG to most cells was highly Zn^{2+} -dependent, whereas binding of the N1N2 domain was unaffected by physiological concentrations of Zn^{2+} . Divalent cations (in particular, Zn^{2+}) are known to interact with the HRR of HRG (3, 4, 35). Kazama and Koide (36) first noted that physiological concentrations of Zn^{2+} (~20 μ M) could enhance the ability of HRG to bind and neutralize heparin in an *in vitro* system. Olsen *et al.* (18) then investigated the effect of Zn^{2+} on HRG cell-surface binding and showed that Zn^{2+} strongly potentiates the binding of HRG to various T-cell lines. Similarly, in this study, we found that physiological concentrations of Zn^{2+} (20 μ M) potentiated the binding of HRG to five of the six cell lines tested, in all cases resulting in at least an ~10-fold increase in HRG binding. The one exception was

human umbilical vein endothelial cells, which failed to interact with HRG in either the presence or absence of Zn^{2+} , indicating that HRG does not bind to heparan sulfate proteoglycans on all cell types. Whether this is due to HRG recognizing a specific heparan sulfate sequence that is absent on endothelial cells or to the requirement for a second receptor remains to be determined. Based on our findings, we propose a model whereby full-length HRG binds with low affinity to cell-surface heparan sulfate in the absence of physiological concentrations of Zn^{2+} via its N1N2 domain. Following chelation of Zn^{2+} by the HRR, however, a change is induced in the molecule that results in the protein binding with higher affinity to cell-surface heparan sulfate. This enhanced binding could be due to the interaction of Zn^{2+} with the HRR, inducing a conformation change within the protein that results in enhanced N1N2 binding to heparan sulfate. Alternatively, Zn^{2+} could mediate cross-linking of HRRs in adjoining HRG molecules, with the resultant dimeric/multimeric HRG complexes having an increased avidity for cell-surface heparan sulfate.

Collectively, this study provides new insights into the cell surface-binding characteristics of HRG. Physiologically, HRG binding to cell-surface heparan sulfate may be regulated by high local Zn^{2+} concentrations that can occur at sites of tissue injury, where degranulating platelets locally release high levels of Zn^{2+} from intracellular stores (37, 38). The modular structure of HRG could provide a mechanism for the molecule to bind to cells via its N1N2 domain and then cross-link other ligands such as plasminogen to cell surfaces via its C-terminal domain. Thus, HRG could act as an adaptor molecule, tethering ligands to cells at sites of high local concentrations of Zn^{2+} . Armed with this important basic understanding of the interaction of HRG with cell surfaces, further studies aimed at identifying the functional role of HRG as an extracellular adaptor molecule may now be pursued.

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